



An experimental and kinetic modeling study of *n*-octane and 2-methylheptane in an opposed-flow diffusion flame

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ABSTRACT

Fischer–Tropsch (FT) fuels derived from biomass syngas are renewable fuels that can replace conventional petroleum fuels in jet engine and diesel engine applications. FT fuels typically contain a high concentration of lightly methylated *iso*-alkanes, whereas petroleum derived jet and diesel fuels contain large fractions of *n*-alkanes, cycloalkanes, and aromatics plus some lightly methylated *iso*-alkanes. In order to better understand the combustion characteristics of FT and petroleum fuels, this study presents new experimental data for 2-methylheptane and *n*-octane in an opposed-flow diffusion flame. The high temperature oxidation of 2-methylheptane and *n*-octane has been modeled using an extended transport database and a reaction mechanism consisting of 3401 reactions involving 714 species. The proposed model shows good qualitative and quantitative agreement with the experimental data. The measured and predicted concentrations of 1-alkenes and ethylene are higher in the *n*-octane flame, while the concentrations of *iso*-alkenes (especially *iso*-butene) and propene are higher in the 2-methylheptane flame. The proposed chemical kinetic model is used to delineate the reactions pathways leading to these observed differences in product species concentrations. An uncertainty analysis was conducted to assess experimental and modeling uncertainties. The results indicate that the simulations are sensitive to the transport parameters used to calculate fuel diffusion.

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1. Introduction

The increasing demand for gasoline, diesel, and jet fuels has put significant pressure on the supply of conventional petroleum feedstock. Therefore, the production of transportation fuels from alternative feedstock is necessary for energy security. Fischer–Tropsch (FT) liquid fuels are one class of fuels that can be produced from other hydrocarbon sources, such as coal and natural gas. The commercialization of FT fuels began in the 1930s, but only in the last 15 years has there been a significant investment into new production facilities [1]. Since FT fuels can be generated from any syngas ($\text{CO} + \text{H}_2$) source [2], the use of gasified biomass to produce FT fuels provides the opportunity for a renewable fuel with significant environmental benefits.

The production of FT fuels initially creates long chain *n*-alkanes (i.e., paraffinic waxes), some of which are partially isomerized to meet jet fuel density and freeze point specifications [2]. Detailed chemical analysis of a natural gas derived FT fuel known as S-8 indicates a mixture of C_7 – C_{18} linear alkanes and *iso*-alkanes with

one or two methyl branches [3,4]. Gokulakrishnan et al. [5] have reported the high temperature ignition characteristics of S-8. Huber et al. [6] have proposed an S-8 surrogate that well produces the thermophysical properties of the real fuel; two of the principal components in the surrogate are 3-methyldecane and 2,6-dimethyloctane. Additionally, Colket et al. [7] proposed 2-methylundecane as a surrogate for the *iso*-alkanes in jet fuel.

The goal of the present study is to better understand the combustion properties of lightly branched alkanes. The focus is on 2-methylheptane (C_8H_{18} -2, refer to Fig. 1), since this is the simplest *iso*-alkane present in S-8 [3,4]. This paper presents a chemical kinetic model for the high temperature combustion of 2-methylheptane and validates it against opposed-flow diffusion flame experiments. Experimental and modeling data for *n*-octane is also included to understand the key differences in the combustion of straight alkanes and 2-methylalkanes. The current work is the first step towards a detailed high temperature and low temperature chemical kinetic model for C_7 – C_{20} 2-methylalkanes.

1.1. Combustion research on 2-methylalkanes

There has been limited research on the fundamental combustion properties of 2-methylalkanes. The smallest 2-methylalkane,

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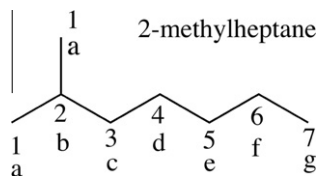


Fig. 1. Structure of 2-methylheptane.

iso-butane (2-methylpropane, iC_4H_{10}), was studied by Wilk et al. [8] in an internal combustion engine, and the results were modeled using a detailed chemical kinetic mechanism. At a given pre-ignition temperature, *iso*-butane is shown to be less reactive than *n*-butane because *iso*-butyl radicals cannot undergo rapid RO_2 isomerization reactions that lead to chain branching. Ogura et al. [9] studied *iso*-butane in a shock tube at high temperatures of 1200–1600 K, 1.7–2.4 atm, and an equivalence ratio of 0.72. They concluded that *iso*-butane has longer ignition delay times than *n*-butane because of the greater amount of propene production, which leads to the resonantly stabilized allyl (C_3H_5) radical. Oehlschlaeger et al. [10] provided shock tube measurements and OH concentration time histories for *iso*-butane and *iso*-pentane (2-methylbutane, iC_5H_{12}) spanning 1117–2009 K, 1.10–12.58 atm, and equivalence ratios of 0.25–2. Recently, Healy et al. [11] presented ignition delay times and an improved chemical kinetic model for *iso*-butane spanning 590–1567 K, 1–30 atm, and equivalence ratios of 0.3–2. For other *iso*-alkanes, Burcat et al. [12] found that the high temperature ignitions kinetics of 2-methylpentane (C_6H_{14-2}) were similar to *n*-hexane at 1175–1772 K and 2–4.6 atm. Their chemical kinetic model indicates that unimolecular decomposition of 2-methylpentane into *n*-propyl (nC_3H_7) and *iso*-propyl (iC_3H_7) is the key initiation step during ignition at high temperature (e.g., 1500 K).

Several experimental and kinetic modeling studies have been performed on 2-methylhexane (C_7H_{16-2}). Westbrook et al. [13] proposed a high temperature model for all nine isomers of heptane, including 2-methylhexane. No experimental results were available for model validation; however, high temperature shock tube simulations indicate that 2-methylhexane has similar ignition delay times to *n*-heptane. H-atom abstraction from the fuel is the primary initiating step, except at the highest temperatures where unimolecular decomposition becomes important. A subsequent study by Westbrook et al. [14] presented a detailed chemical mechanism including low temperature chemistry for 2-methylhexane and used rapid compression machine data from Griffiths et al. [15] for validation. Their simulations indicate that 2-methylhexane is noticeably slower to react than *n*-heptane in the low and intermediate temperature regimes (e.g., 650–900 K). Silke et al. [16] also reported that 2-methylhexane is less reactive than *n*-heptane in rapid compression experiments from 650 to 950 K at 15 atm.

Currently, the largest 2-methylalkane to be studied is 2-methylheptane (C_8H_{18-2}). Kahandawala et al. [17] reported high temperature shock tube ignition delays and particulate matter (PM) emissions for jet fuel (JP-8), F-T jet fuel (S-8), *n*-heptane/toluene mixture (surrogate JP-8), and 2-methylheptane (surrogate S-8). They found that both S-8 and 2-methylheptane produce less PM emissions than the real and surrogate JP-8. However, no differences in ignition delay times were observed for both actual and surrogate JP-8 and S-8 at the high temperatures studied (i.e., 1100–1600 K).

2. Experimental methods

2.1. Opposed-flow diffusion flame

The experimental setup is similar to that described earlier by Sarathy et al. [18]. The setup consists of two identical flat flame

burners with circular burner ports of diameter 25.4 mm, facing each other and spaced 20 mm apart. A fuel mixture of 98.14% N_2 and 1.86% fuel (99% pure *n*-octane or 98% pure 2-methylheptane) is fed through the bottom port at a mass flux rate of 0.015 g/cm² s, while an oxidizer mixture of 42.25% O_2 and 57.75% N_2 is fed through the top port at a mass flux rate of 0.014 g/cm² s. At these plug flow conditions, the Reynold's Number at the fuel port exit is in the laminar flow regime (i.e., $Re = 176$), the flame is on the fuel side of the stagnation plane, and the fuel side strain rate is approximately $36\ s^{-1}$. To prepare the fuel mixture, a peristaltic pump delivered the fuel to an ultrasonic atomizer which sprays the liquid fuel into a stream of N_2 gas. The gaseous fuel mixture is delivered to the burner via heated stainless steel tubing. The temperatures of the gases exiting the top and bottom burner ports were 420 K and 350 K, respectively. The mass flow controllers used to meter the delivery of nitrogen, oxygen, and air have a manufacturer reported error of $\pm 1\%$. The pump used to deliver the liquid fuel was calibrated before each experiment to an accuracy of $\pm 2\%$.

The gas sampling system in these experiments consists of a fused-silica microprobe (0.20 mm internal diameter, 0.32 mm outer diameter, and 5 cm length) connected to a dual-stage pump with heated heads (420 K) and PTFE diaphragms to prevent the condensation of high boiling point compounds. The suction side of the sampling system consists of heated stainless steel tubing (6.35 mm internal diameter and 2 m length) and a pressure gauge connected to the quartz microprobe. An absolute pressure of 4–6 kPa was measured downstream of the microprobe. This is sufficient to quench most reactions and ensure accurate data on flame composition. The compression side of the pump delivers the samples to the analytical instruments via heated stainless steel tubing (6.35 mm internal diameter and 2 m length). The temperature of all heated surfaces is 400 K, which is sufficient to avoid condensation of high molecular weight compounds (e.g., the fuel) in the sampling lines.

Analytical techniques used to analyze the species in the sample includes: GC/FID with an HP-AL/S PLOT column for C_1 – C_8 hydrocarbons and GC/TCD for CO and CO_2 . Temperature measurements were obtained using both a 100 μm and a 250 μm wire diameter R-type thermocouple (uncoated Pt/Pt-13%Rh) in an apparatus similar to that used by McEnally et al. [19]. This apparatus was unable to accurately measure temperatures on the oxidizer-rich side of the flame because of heating of the thermocouple wires by exhaust products from the flame. Therefore, measurements above 11 mm from the fuel port are not reported. The measured temperatures were corrected for radiation losses using the method described by Shaddix [20]. The Nusselt number was calculated assuming a spherical bead of the same diameter as the wires and Prandtl numbers for pure N_2 at 398 K. The emissivity was obtained from [21] and the thermal conductivity of air was from [22]. The possibility of catalytic reactions occurring on the uncoated thermocouple wires is acknowledged [20], but were not corrected. Fluctuations of ± 50 –80 K were observed in the hot regions near the flame front (i.e., above 1300 K), so we report the 10 s average. We estimate a maximum error of $\pm 10\%$ in the temperature measurements, and the precision of species measurements is estimated to be $\pm 15\%$.

3. Computational methods

The kinetic modeling for 2-methylheptane and *n*-octane was performed using the CHEMKIN PRO modeling package [23]. The opposed-flow diffusion flame was modeled using the OPPDIF code. The reported simulations used mixture-averaged transport and included thermal diffusion. It should be noted that the computationally expensive multi-component transport method was also tested, and the predicted maximum concentrations were similar

to the mixture-averaged predictions. The solution was assumed grid-independent when increasing the number of grid points did not change the simulated temperature and species profiles (i.e., 150 grid points).

The data files for the simulation include a detailed chemical kinetic reaction mechanism, a dataset of thermochemical properties, and a dataset of transport properties. The entire model consists of 3401 reactions involving 714 species. These input files are available as [Supplemental material](https://www-pls.llnl.gov/?url=science_and_technology-chemistry-combustion) to this publication and from our website at: https://www-pls.llnl.gov/?url=science_and_technology-chemistry-combustion.

3.1. Chemical kinetic mechanism

The chemical kinetic reaction mechanism for 2-methylheptane builds upon previous *n*-alkane [24] and *iso*-octane mechanisms [25,26]. The base mechanism consists of comprehensive low temperature and high temperature reactions for C₀–C₇ species from [25,26] plus high temperature reactions for *n*-octane from [24]. The present mechanism was then built in a modular fashion, starting with the high temperature reactions for 2-methylhexane described by Westbrook et al. [13]. The high temperature reactions for 2-methylheptane were then written for each of the nine classes described by Curran et al. [27].

1. Unimolecular decomposition.
2. H-atom abstraction from the fuel.
3. Alkyl radical decomposition.
4. Alkyl radical + O₂ to produce alkene and HO₂ directly.
5. Alkyl radical isomerization.
6. Abstraction reactions from alkenes by OH, H, O, and CH₃.
7. Addition of radical species to alkenes.
8. Alkenyl radical decomposition.
9. Alkene decomposition.

2-Methylheptane is denoted as C₈H₁₈-2 in the mechanism (see Fig. 1). The carbon chain is labeled numerically (i.e., 1, 2, 3, etc.) such that the location number of the methyl branch is minimized. For 2-methylheptene species, the location of a double bond is identified by a hyphen followed by the number of the first carbon in the double bond (e.g., 2-methyl-3-heptene is C₇H₁₄-3-2). Additional notations are provided to denote radical sites in the molecule. The carbon sites are labeled alphabetically (i.e., a, b, c, etc.) such that the location of the first methyl branch is minimized (see Fig. 1). In this way, the 2-methyl-3-heptyl radical is denoted as C₇H₁₅-2c, while the 2-methyl-1-heptyl radical is written as C₇H₁₅-2a.

The reaction rate rules used for reaction classes 1–4 and 6–9 are the same as those in the *iso*-octane mechanisms [25,26,28]. For example, H-atom abstraction by OH from the tertiary carbon in 2-methylheptane has the same rate as the analogous reaction in *iso*-octane (i.e., abstraction from a tertiary carbon in *iso*-octane). Unimolecular decomposition reactions are specified in the reverse direction (i.e., recombination), and the rate of recombining two radicals is assigned. The decomposition rate is computed using the principle of microscopic reversibility. Retroene reactions for 2-methylalkenes were added to the conventional class 9 reactions using the rate expressions from [24].

For both *n*-octane and 2-methylheptane, the class 5 radical isomerizations were written to include reactions involving 5-member, 6-member, and 7-member transition state rings, and their rates were updated based on recent kinetic studies. We did not include radical isomerizations involving fewer than five and greater than seven members because they are much slower. The rates for primary to secondary radical isomerizations involving 5- and 6-members were obtained from the work of Tsang et al. on hexyl radicals [29]. The rate for primary to primary isomerization

involving 7-members and the rates for secondary to secondary isomerizations involving 5- and 6-members were obtained from the work of Tsang et al. on octyl radicals [30]. The rate for tertiary to primary isomerization involving 5-members was obtained from the work of McGivern et al. on 2-methylpentyl radicals [31]. Finally, the rates for tertiary to primary involving 6-members and tertiary to secondary isomerizations involving 5- and 6-members were obtained from the recent work of Awan et al. on 2-methylhexyl radicals [32]. We could not find a published rate for the tertiary to primary isomerization involving 7-members, so we estimated the rate at 1000 K to be approximately 1.5 times slower than the same isomerization involving 6-members, and then developed a rate constant to fit this. This is a reasonable assumption since Tsang et al. [30] showed for *n*-octyl that the primary to secondary isomerization involving 7-members is 1.5 times slower at 1000 K the same isomerization involving 6-members.

3.2. Thermochemical data

The THERM [33] software was used to compute the thermochemical properties of species not present in the *n*-alkane model [24]. The THERM group values are from Benson [34] and Bozzelli et al. [35].

3.3. Transport properties

In certain combustion applications, such as shock tubes, the overall rate is assumed to be kinetically controlled since fuel and air are premixed and combustion is assumed to occur homogeneously. Therefore, many published models exclude transport property databases. However, both kinetic processes and transport processes are rate controlling in laminar diffusion flames, so transport properties are needed for all the species in the model. This study obtained the molecular transport parameters for species using a variety of methods. The transport properties for the majority of compounds were already available in a previously published primary reference fuel (PRF) model [24,26]. The Lennard-Jones (LJ) collision diameter and potential well depth were changed for a number of small species based on the recommendations of Mourits and Rummens [36]. The transport data files contains comments to indicate which species were modified. The transport properties of larger alkane, alkene, alkyl, and alkenyl species were determined as follows. For stable species (e.g., *n*-octane and 2-methylheptane), this study used the correlations developed by Tee et al. [37], as described in Holley and coworkers for hydrocarbons [38], to calculate the LJ collision diameter and potential well depth using the critical pressure (*P*_c), critical temperature (*T*_c), and boiling point (*T*_b) of the species. *P*_c, *T*_c, and *T*_b for stable species were obtained from [38]. The polarizability in cubic Angstroms of stable species was obtained from experimentally measured values available in [22]. The dipole moment was obtained from [40]. The index factor which describes the geometry of the molecule was determined from the molecular structure (i.e., 0 for atoms, 1 for linear molecules, and 2 for nonlinear molecules). We assumed that the transport properties are similar for alkanes and alkenes of the same chain length. For alkyl and alkenyl radical species, the transport properties of their stable counterpart were used.

4. Results and discussion

The proposed chemical kinetic model was validated against experimental data for *n*-octane and 2-methylheptane in an opposed-flow diffusion flame. The opposed-flow diffusion flame allows us to study the oxidation of these hydrocarbon fuels in a non-premixed flame environment. Temperature and species

concentration profiles were obtained by sampling and measuring the product gas at various points between the two burner ports. The measured species included *n*-octane ($n\text{-C}_8\text{H}_{18}$), 2-methylheptane ($\text{C}_8\text{H}_{18-2}$), carbon monoxide (CO), carbon dioxide (CO_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), acetylene (C_2H_2), propene (C_3H_6), *iso*-butene ($i\text{-C}_4\text{H}_8$), 1-pentene ($\text{C}_5\text{H}_{10-1}$), 3-methyl-1-butene (denoted as 2-methyl-3-butene, $\text{C}_5\text{H}_{10-3-2}$), 1-hexene ($\text{C}_6\text{H}_{12-1}$), and 4-methyl-1-pentene (denoted in the mechanism as 2-methyl-4-pentene, $\text{C}_6\text{H}_{12-4-2}$). The gas chromatography method could not separate 1-butene ($1\text{-C}_4\text{H}_8$) from 1,2-propadiene ($1,2\text{-C}_3\text{H}_4$), so we report the combined concentration of these two species. The same was done for 1,3-butadiene ($1,3\text{-C}_4\text{H}_6$) and propyne (C_3H_4). We did not attempt to measure oxygenated species such as aldehydes and ketones. Besides the alkenes mentioned above, the analytical system was not designed to measure the isomers of pentene, hexene, heptene, and octene. Species below the experimental limit of quantification (5 PPM) included propane, *n*-butane, 1-butyne, 2-butyne, *trans*-2-butene, *cis*-2-butene, *n*-pentane, and *n*-hexane.

The reported species and temperature profiles were shifted by 0.5 mm away from the fuel port to account for a positioning uncertainty, which is a human error introduced when zeroing the sampling probe against the fuel port surface. It is often acceptable to arbitrarily shift species profiles [41] to account for probe-induced or thermocouple-induced perturbations to the flame structure. These perturbations include flow-field disturbances and cooling effects caused by introducing a foreign object into the flame. The profiles reported in this study have not been shifted to account for such perturbations because we do not have a quantitative analysis of their contributions to flame structure disturbance.

4.1. 2-Methylheptane results

Figure 2 displays the measured and predicted species and temperature profiles obtained in the 2-methylheptane opposed-flow diffusion flame. $\text{C}_6\text{H}_{12-4-2}$ is not plotted since both peak measured and predicted concentrations are below 100 PPM. The experimental errors in species concentrations are too high for concentrations below 1000 PPM. The experimental results (solid symbols) show that the 2-methylheptane concentration decreases quickly at a distance of 5.5 mm from the fuel port. As the fuel is consumed, the CO and CO_2 concentrations begin rising. All of the fuel is consumed at a distance of approximately 8.25 mm from the fuel port, which corresponds closely to the peak measured temperature (1688 K). The flame front was visually observed at approximately 8.5 mm from the fuel port. Just before the flame front, at around 8 mm from the fuel port, the concentrations of hydrocarbon species reach their peak. Besides CO and CO_2 , the most abundant measured species are C_2H_4 , C_2H_2 , CH_4 , and C_3H_6 .

The model well reproduces the shape of the experimental temperature and species profiles; however, the experimental data is shifted by about 0.5 mm towards the fuel port. This discrepancy could be resolved by arbitrarily shifting the experimental profiles, which is widely accepted [41]. In the following discussion, the model's quantitative prediction is considered good if the predicted maximum mole fraction is within a factor 1.5 of the measured maximum mole fraction. The model well reproduces the experimentally measured temperature profile, but the maximum measured temperature is more than 100 K higher than the predicted value. A delay (i.e., shift away from the fuel port) in the predicted reactivity of 2-methylheptane is observed. The maximum concentration of CO_2 is well predicted, while the maximum concentration of CO is under predicted by approximately 1.4 times. The model performs well at predicting the peak measured concentrations of CH_4 , C_2H_4 , C_2H_2 , C_2H_6 , $1,3\text{-C}_4\text{H}_6 + \text{C}_3\text{H}_4$, $i\text{-C}_4\text{H}_8$, $\text{C}_5\text{H}_{10-1}$, and $\text{C}_5\text{H}_{10-3-2}$, and $1\text{-C}_4\text{H}_8 + 1,2\text{-C}_3\text{H}_4$. The peak concentration of C_3H_6 is under predicted by nearly 1.5 times.

4.2. *n*-Octane results

Figure 3 displays the measured and predicted species and temperature profiles obtained in the *n*-octane opposed-flow diffusion flame. The experimental fuel, CO, CO_2 , and temperature profiles are comparable to those of 2-methylheptane. Therefore, both fuels have similar reactivity under the present conditions.

The proposed model well reproduces the shape of the experimental temperature and species profiles, with the same 0.5 mm shift observed as for 2-methylheptane. The model well reproduces the experimentally measured temperature profile, the profile for *n*-octane, and the peak concentration of CO_2 . The peak concentration of CO is under predicted by approximately 1.4 times. The model also well predicts the peak measured concentrations of CH_4 , C_2H_4 , C_2H_2 , C_2H_6 , $1,3\text{-C}_4\text{H}_6 + \text{C}_3\text{H}_4$, $\text{C}_5\text{H}_{10-1}$, $\text{C}_6\text{H}_{12-1}$ and $1\text{-C}_4\text{H}_8 + 1,2\text{-C}_3\text{H}_4$. The maximum concentration of C_3H_6 is under predicted by approximately 1.7 times.

4.3. Discussion of 2-methylheptane and *n*-octane results

Experiments and simulations were conducted to elucidate the differences in combustion between an *n*-alkane and a 2-methylalkane. Table 1 presents the peak measured and predicted mole fractions in the *n*-octane and 2-methylheptane flames. Considering that the experimental error is $\pm 15\%$, the two flames have similar concentrations of CO_2 , CO, CH_4 , C_2H_6 , and C_2H_2 . Both the experiments and simulations indicate that the 2-methylheptane flame produces more C_3H_6 and 2-methylalkenes (e.g., $i\text{-C}_4\text{H}_8$, $\text{C}_5\text{H}_{10-3-2}$, and $\text{C}_6\text{H}_{12-4-2}$) while the *n*-octane flame produces more C_2H_4 and 1-alkenes (e.g., $\text{C}_5\text{H}_{10-1}$ and $\text{C}_6\text{H}_{12-1}$). The experimental data does not allow us individually quantify $1\text{-C}_4\text{H}_8$, $1,2\text{-C}_3\text{H}_4$, $1,3\text{-C}_4\text{H}_6$, and C_3H_4 ; however, the model predicts that the *n*-octane flame produces more $1\text{-C}_4\text{H}_8$ and $1,3\text{-C}_4\text{H}_6$, while the 2-methylheptane flame produces more $1,2\text{-C}_3\text{H}_4$ and C_3H_4 .

Reaction path analyses were performed using the proposed model to explain the difference observed for the *n*-octane and 2-methylheptane flames. Reaction path analyses were performed at two different temperatures, as follows:

- Low temperature, $T = 805$ K, corresponding to approx. 7.2 mm from the fuel port and 21% of the fuel consumed.
- Intermediate temperature, $T = 1185$ K, corresponding to 8.1 mm from the fuel port and 80% of the fuel consumed. These conditions also correspond closely to the peak of many hydrocarbon products.

Figure 4 displays the primary reactions paths involved in the consumption of 2-methylheptane at the two aforementioned temperatures, with italic and bold texts referring to low and intermediate temperature conditions, respectively. Percentages corresponding to each pathway are rounded to the nearest whole number. The low temperature analysis provides an understanding of how fuel consumption initiates in the flame. According to the proposed model, H-atom abstraction accounts for the nearly 100% of the fuel consumption, with abstraction being dominated by H atoms, OH radicals, and CH_3 radicals. At intermediate temperatures, the analysis provides an understanding of the reactions occurring near the peak of hydrocarbon products. Again, the fuel consumption is dominated by H-atom abstraction reactions by H atoms, OH radicals, and CH_3 radicals. At high temperatures (e.g., 1500 K, not shown in Fig. 4), H-atom abstraction reactions are negligible and the energy is now available to activate unimolecular decomposition reactions. However, the amount of fuel that reaches these conditions is small, so these reactions do not contribute significantly to the formation of product species.

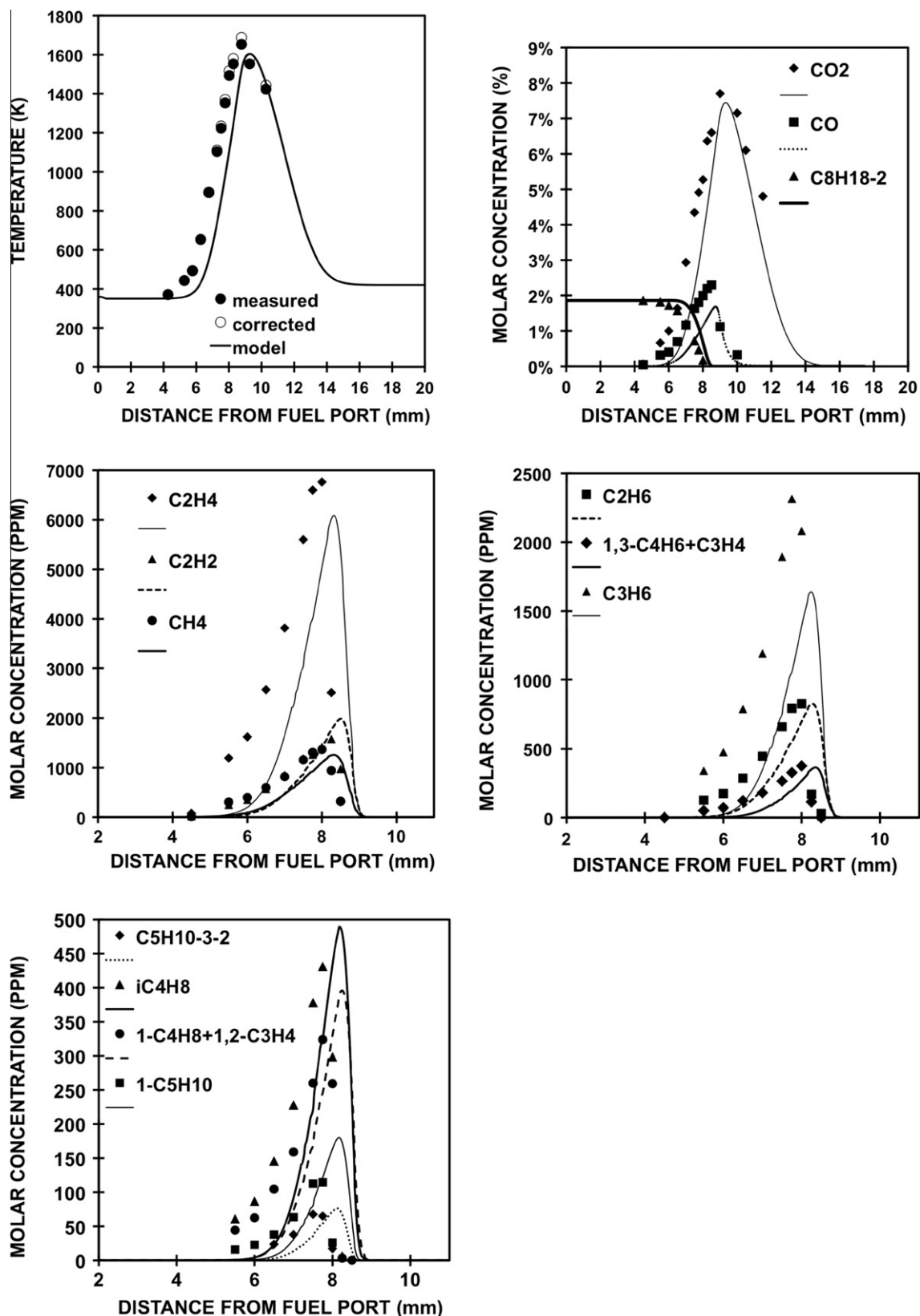


Fig. 2. Experimental and computed profiles obtained from the oxidation of 2-methylheptane in an atmospheric opposed-flow flame (1.86% fuel, 42% O_2).

At both low and high temperatures, H-atom abstraction from the tertiary carbon is the most predominant route (20%) because tertiary C–H bonds are weaker than secondary and primary C–H bonds. The abstraction of secondary H atoms is equally distributed

amongst the four secondary carbon sites (i.e., 16% each). There are three primary carbon atoms in 2-methylheptane and 5% of the fuel is consumed via H-atom abstraction from each of these sites; however, H-atom abstraction from the two primary carbon atoms near

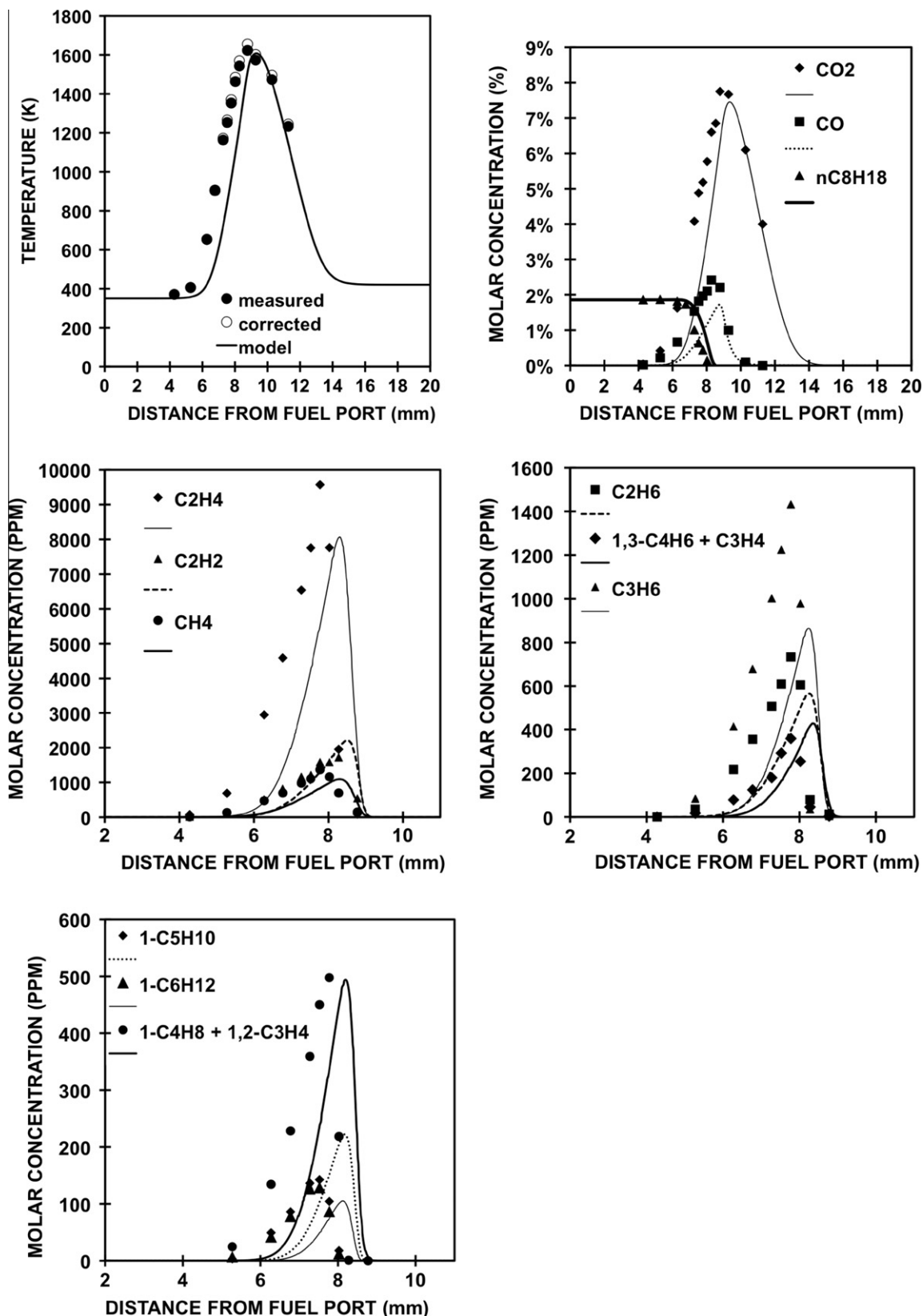


Fig. 3. Experimental and computed profiles obtained from the oxidation of *n*-octane in an atmospheric opposed-flow flame (1.86% fuel, 42% O₂).

the methyl branch produces the same radical, so this route is effectively doubled (i.e., 10% leads to C₈H₁₇-2a). Figure 4 does not display the fate of the C₈H₁₇-2 g since only 5% of the fuel produces this radical. Figure 4 indicates that the fuel radicals formed via

H-atom abstraction are primarily consumed via β -scission. However, radical isomerization reactions are also important; most notably the secondary to tertiary radical isomerizations involving 5- and 6-membered transition state rings and primary to tertiary

Table 1

Peak measured and predicted temperatures and concentrations in the opposed-flow diffusion flame. Italicized numbers are measured values, bold numbers represent predicted values, and underlined numbers are the ratio of measured to predicted.

Measured parameter	$n\text{-C}_8\text{H}_{18}/$ $n\text{-octane}$	$\text{C}_8\text{H}_{18}\text{-2}/$ 2-methylheptane
Temperature (K)	1655 1606 <u>1.0</u>	1688 1604 <u>1.0</u>
CO ₂ /carbon dioxide (%)	7.75 7.46 <u>1.0</u>	7.70 7.45 <u>1.0</u>
CO/carbon monoxide (%)	2.42 1.72 <u>1.4</u>	2.30 1.70 <u>1.4</u>
CH ₄ /methane (PPM)	1369 1096 <u>1.2</u>	1368 1261 <u>1.1</u>
C ₂ H ₆ /ethane (PPM)	734 566 <u>1.3</u>	827 833 <u>1.0</u>
C ₂ H ₄ /ethylene (PPM)	9571 8068 <u>1.2</u>	6763 6146 <u>1.1</u>
C ₂ H ₂ /acetylene (PPM)	1582 2215 <u>0.7</u>	1575 1987 <u>0.8</u>
C ₃ H ₆ /propene (PPM)	1432 865 <u>1.7</u>	2314 1646 <u>1.4</u>
1-C ₄ H ₈ + 1,2-C ₃ H ₄ /1-butene + 1,2-propadiene (PPM)	498 494 <u>1.0</u>	324 400 <u>0.81</u>
1,3-C ₄ H ₆ + C ₃ H ₄ /1,3-butadiene + propyne (PPM)	359 428 <u>0.8</u>	376 358 <u>1.1</u>
iC ₄ H ₈ /iso-butene (PPM)	<LOD 2 <u>N/A</u>	431 480 <u>0.9</u>
1-C ₅ H ₁₀ /1-pentene (PPM)	142 222 <u>0.6</u>	114 163 <u>0.4</u>
C ₅ H ₁₀ -3-2/2-methyl-3-butene (PPM) aka 3-methyl-1-butene	<LOD 0 <u>N/A</u>	68 75 <u>0.9</u>
1-C ₆ H ₁₂ /1-hexene (PPM)	129 105 <u>1.2</u>	<LOD 1 <u>N/A</u>
C ₆ H ₁₂ -4-2/2-methyl-4-pentene (PPM) aka 4-methyl-1-pentene	<LOD 0 <u>N/A</u>	40 30 <u>1.3</u>

radical isomerization involving 7-members (not shown in Figure). These reactions are important because the tertiary C–H bond is weaker than secondary and primary C–H bonds. The radical formed (i.e., C₈H₁₇-2a) after abstraction from the tertiary site eventually leads to the formation of iC₄H₈ and C₂H₄. Fuel radicals with a radical site on a secondary carbon atom can undergo β-scission to form either a 1-alkene or a 2-methylalkene. However, β-scission reactions tend not to favor the formation of a methyl radical, so the routes leading to C₃H₆, C₄H₈-1, and C₅H₁₀-3-2 are predominant. Ethyl (C₂H₅) and *n*-propyl (C₃H₇) appear in a number of pathways, and these eventually lead to the formation of C₂H₄ and C₃H₆.

Figure 5 displays the reactions paths involved in the consumption of *n*-octane. *n*-Octane is a symmetric molecule, so this figure displays pathways for the four possible *n*-octyl radicals. Therefore, the percentages displayed for each pathway are effectively doubled. At both low and intermediate temperatures, H-atom abstraction accounts for the nearly all of the fuel consumption, with abstraction being dominated by H atoms, OH radicals, and CH₃ radicals.

H-atom abstraction from the secondary carbon atoms is the most predominant route because secondary C–H bonds are weaker than primary C–H bonds. The abstraction of secondary H atoms is equally distributed amongst the six secondary carbon sites (i.e., 15% each). H-atom abstraction from the two primary carbon atoms accounts for 9% of the fuel consumption (i.e., 4.5% each). The fuel radicals formed are primarily consumed via β-scission reactions. However, at low temperatures some of the C₈H₁₇-1 radical undergoes isomerization to form C₈H₁₇-4 via 5-membered and 6-membered transition states. Similarly, the C₈H₁₇-2 radical can undergo isomerization to form C₈H₁₇-3 via a 6-membered transition state at low temperatures.

β-scission reactions produce C₄H₈-1, C₅H₁₀-1, and C₆H₁₂-1 for fuel radicals with a radical site on a secondary carbon. C₂H₄ also appears to be formed in nearly every pathway, either directly or via ethyl (C₂H₅) and *n*-propyl (C₃H₇).

As mentioned previously, the 2-methylheptane flame produces more C₃H₆, iC₄H₈, C₅H₁₀-3-2, and C₆H₁₂-4-2, and the *n*-octane flame produces more C₂H₄, 1-C₄H₈, C₅H₁₀-1 and C₆H₁₂-1. The model's reaction pathways clearly display how the combustion of 2-methylheptane leads to significant amounts of propene and *iso*-alkenes, while *n*-octane leads to the formation of ethylene and 1-alkenes. The simulations predict that the 2-methylheptane flame produces more 1,2-C₃H₄ (allene) and C₃H₄ (propyne). These species are produced when an H-atom is abstracted from C₃H₆, and the subsequent C₃H₅ radical undergoes β-scission. Allene and propyne are formed in higher concentrations in the 2-methylheptane flame because the C₃H₆ concentration is greater than in the *n*-octane flame. On the other hand, the simulations predict higher levels of 1,3-C₄H₆ in the *n*-octane flame, which is formed when an H-atom is abstracted from 1-C₄H₈ and the subsequent C₄H₇ radical undergoes β-scission. Therefore, the higher levels of 1-C₄H₈ in the *n*-octane flame are responsible for the higher concentrations of 1,3-C₄H₆.

5. Uncertainty analysis

This section addresses discrepancies between the experimental data and model predicted values by critically assessing potential sources of error. On the experimental side, errors can be introduced during the supply of reactants to the burner or during sampling. From a modeling perspective, errors can be introduced by missing reaction pathways, uncertainties in reaction rate constants, thermochemical data, or transport parameters.

5.1. Experimental uncertainties

The slight under prediction of peak species concentrations (e.g., CO, C₂H₄, C₃H₆, etc.) and temperature suggests that the carbon balance in the experimental flame is different from that of the simulated flame. The gaseous reactants (air, oxygen, and nitrogen) were introduced to the burner using mass flow controllers with a manufacturer reported error of 1%, so their flows are considered accurate. The fuel was introduced using a peristaltic pump with an estimated error of 2% (e.g., ±0.01 mL/min). Extra precautions were taken to confirm that the delivery of reactants to the burner was accurate. All the supply lines were initially tested for leaks. In

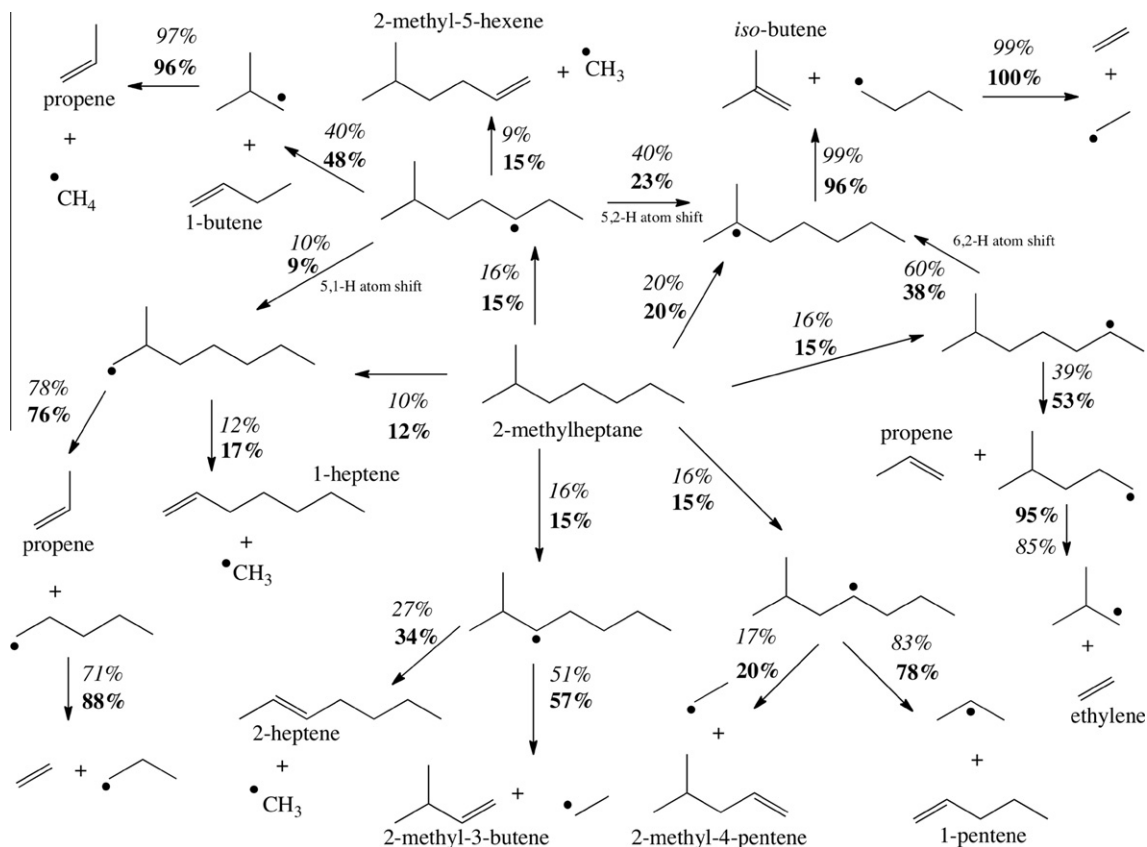


Fig. 4. Reaction pathway diagram for 2-methylheptane oxidation in the opposed-flow diffusion flame at $T = 805$ K (italicized text) and $T = 1185$ K (bold text).

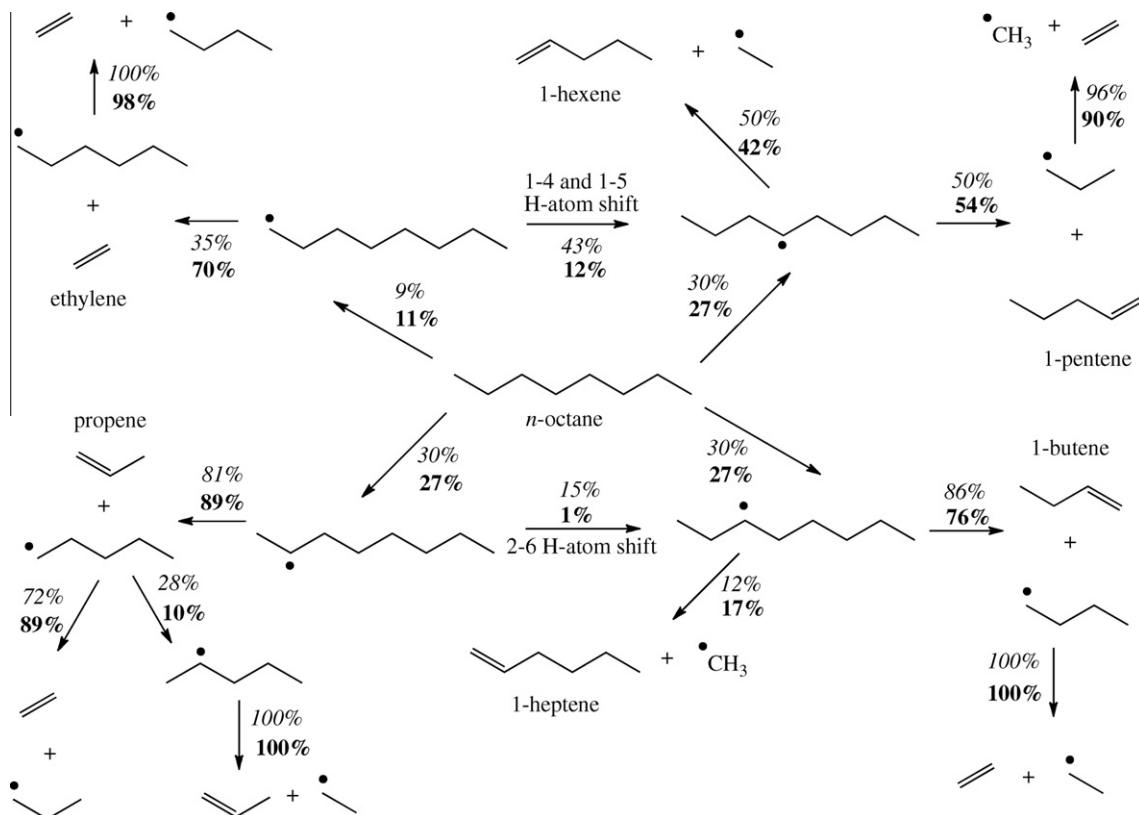


Fig. 5. Reaction pathway diagram for *n*-octane oxidation in the opposed-flow diffusion flame at $T = 805$ K (italicized text) and $T = 1185$ K (bold text).

addition, the fuel supply system was designed and tested to ensure complete vaporization of the fuel, homogeneous mixing with nitrogen, and no fuel condensation in the supply lines. Given these precautions, we are confident in the boundary conditions used for

simulating the experiments. Nevertheless, a flame simulation was conducted assuming a 2% (vol.) increase in the liquid flow rate, which corresponds to an increase in the fuel stream mole fraction from 1.86% to 1.88%. The additional carbon to the flame slightly

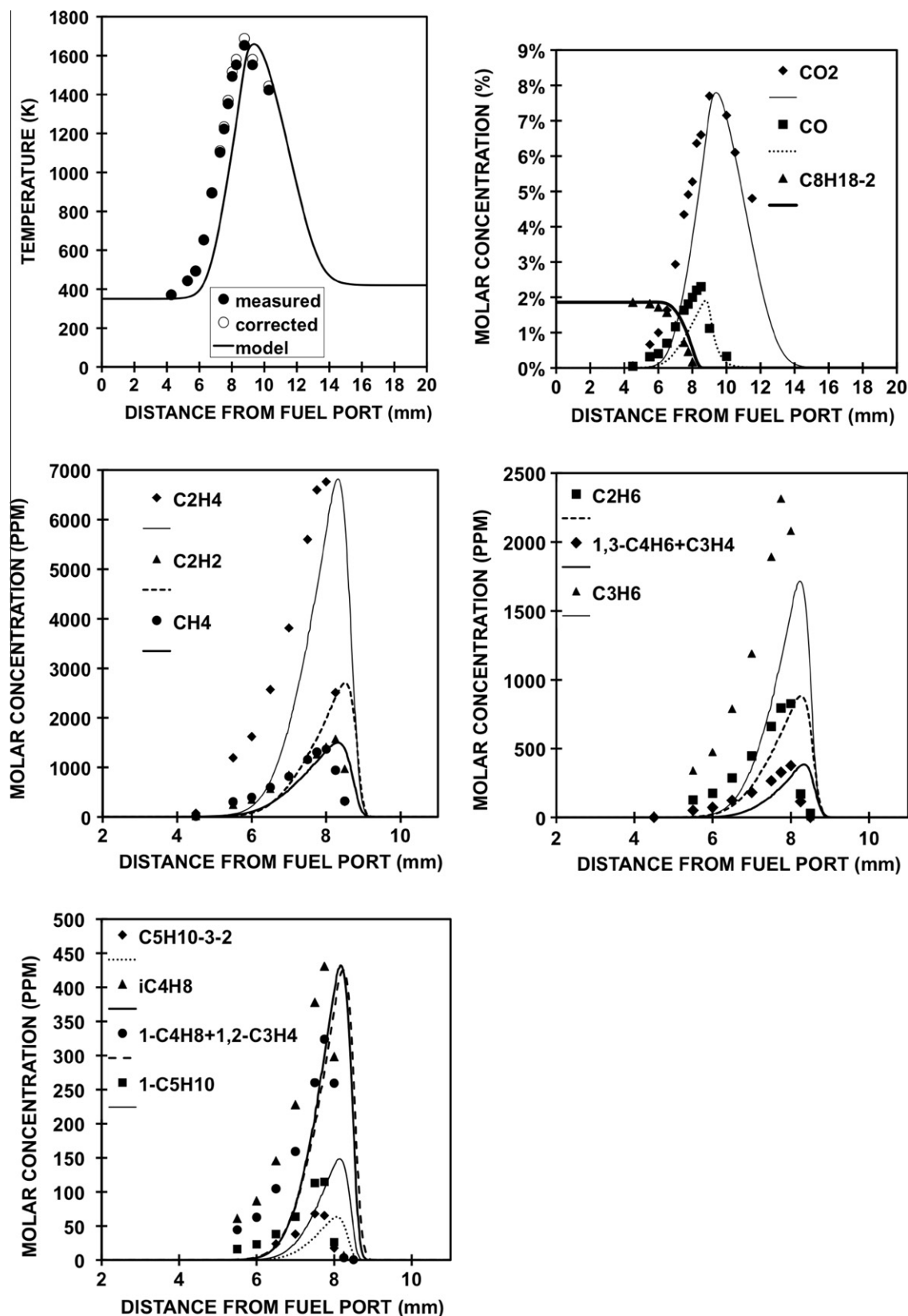


Fig. 6. Experimental and computed profiles obtained from the oxidation of 2-methylheptane in an atmospheric opposed-flow flame (1.86% fuel, 42% O₂) using a 30% reduction in 2-methylheptane collision diameter.

increases the maximum predicted temperature (+30 K) and species concentrations, but the change is small compared to the magnitude of the initial discrepancies.

Sampling errors may also be a source of discrepancies between the model and experiments. It is noted that the difference between measured and predicted temperatures could be due to catalytic reactions on the thermocouple wires [19], but this was not investigated in the present study. The microprobe and sampling line apparatus used for species sampling may introduce errors through flame perturbation and chemical reactions in the sampling line. During the experiments, we did not observe any visual perturbations to the flame sheet by the fused-silica microprobe. Nevertheless, the 0.5 mm shift between the experiments and simulations alludes to experimental error due to flame perturbation. To avoid chemical reactions in the sampling line, we designed the apparatus to allow supersonic expansion of the sample gas by pumping it through a narrow diameter orifice (i.e., 0.2 mm) and into a low pressure environment (i.e., 5 kPa). In this process, the gas is rapidly cooled to 400 K, and the local number density of gas molecules and their collision rate decreases rapidly.

5.2. Modeling uncertainties

Discrepancies between the model and experiments may also be attributed to uncertainties in the kinetic rate constants, thermochemical data, and transport parameters used in the model. Previous research [38–46] in opposed-flow laminar flames indicates that fuel reactivity is more sensitive to diffusion estimates than to kinetic rate parameters because transport processes are rate controlling in these systems. Therefore, tuning rate and thermochemical constants to fit non-premixed flame data is not recommended. The testing and optimization of kinetic parameters is better suited for homogeneous combustion environments such as jet stirred reactors, premixed flames, and shock tube reactors. However, the adjustment of transport parameters to match the reactivity of diffusion-controlled systems is warranted.

Both Holley et al. [38] and Smallbone et al. [45] argue that the molecular diffusion model used in flame simulations is inaccurate because it assumes that the spherical potential of molecules is valid at elevated temperatures. They propose that the LJ transport parameters obtained at low temperature are not expected to accurately predict high temperature diffusion. In addition, Chae and Violi [47] state that the molecular geometry of large *n*-alkanes are non-spherical and better resemble “stick-like” structures. More recently, Chae and Violi [48,49] performed molecular dynamics simulations on heptane isomers to obtain accurate binary diffusion coefficients. Their results suggest that the conventional Chapman–Enskog equation with Lennard–Jones transport parameters derived from correlations of states values do not reproduce binary diffusion coefficients (hydrocarbon:nitrogen) determined using more accurate molecular dynamic simulations. It should be noted that their simulations do not directly address transport in multi-component systems, such as the flames presented in this study. Nevertheless, these uncertainties with respect to transport properties may also play a role in the present study, since the collision diameter for *n*-octane and 2-methylheptane were obtained from empirical correlations.

As a sensitivity analysis, the collision diameter of 2-methylheptane was arbitrarily reduced by 30% from 6.24 to 4.37. The results, shown in Fig. 6, indicate improvements in the agreement between the model predictions and experiments. The predicted fuel profile is shifted towards the fuel port and more closely matches the shape of the experimental profile, which suggests that its transport is better modeled. The peak temperature, CO, CO₂, and minor species profiles are also better predicted by the model. Most notably, the ethylene predictions are improved; however, an over

prediction of predicted acetylene concentration is observed. These results display the sensitivity of predicted species and temperature profiles to the fuel's collision diameter. Further investigation is required to create models that more accurately represent the transport of large fuel molecules in multi-component high temperature diffusion controlled environments.

6. Conclusions

The goal of this study was to improve our understanding of 2-methylheptane combustion. This was achieved by using experimental data for 2-methylheptane and *n*-octane in an opposed-flow diffusion flame to validate a novel high temperature chemical kinetic model. The experimental data indicate that the most abundant measured species were CO, CO₂, C₂H₄, and C₃H₆. Greater quantities of the 1-alkenes and ethylene were detected in the *n*-octane flame, while higher levels of 2-methylalkenes and propene were observed in the 2-methylheptane flame. The proposed chemical kinetic model exhibited excellent agreement with the experimental data. The reaction path analyses performed clearly indicate that H-atom abstraction reactions are the dominant route of fuel consumption. β -scission of the fuel radicals leads to formation of the species observed experimentally. Radical isomerizations are also shown to be important, especially those involving the tertiary site in 2-methylheptane. The model shows that the higher levels of 2-methylalkenes in the 2-methylheptane flame can be attributed to the presence of the methylated portion of the molecule.

The proposed model is the first step towards developing a comprehensive chemical kinetic model for C₇–C₂₀ 2-methylalkanes with detailed low temperature and high temperature chemistry. This study verifies that the pathways and rate rules used for the high temperature reactions are acceptable for modeling the combustion of 2-methylalkanes in diffusion flames.

The uncertainty analysis indicated that predicted species and temperature profiles were found to be sensitive to fuel diffusion, and an arbitrary 30% reduction in the collision diameter was found to improve the agreement between the model predictions and experiments. This study highlights the importance of further experimental and numerical studies to improve our understanding of transport processes in multi-component high temperature diffusion controlled environments.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2010.11.008.

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